APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination

MODEL QUESTION PAPER — SET 2

Course Code: PCCST501 **Course Name:** Computer Networks

Maximum Marks: 60 **Duration:** 2 hours 30 minutes

PART A — Answer all questions. Each question carries 3 marks.

Q. No.	Question	Marks	CO
1	Explain the purpose of encapsulation and decapsulation in protocol layering.	3	CO1
2	Distinguish between HTTP and FTP based on their application-layer functions.	3	CO1
3	Explain the difference between a TCP socket and a UDP socket in terms of communication behaviour.	3	CO2
4	Describe the role of the select() and poll() functions in handling multiple socket descriptors.	3	CO2
5	Explain the purpose of a routing table in forwarding network-layer packets.	3	CO3
6	Differentiate between intra-domain routing and inter-domain routing.	3	CO3
7	Describe the purpose of link-layer addressing in Ethernet communication.	3	CO4
8	Explain the meaning of bandwidth utilization in data communication.	3	CO5

PART B — Answer one full question from each module. Each question carries 9 marks.

Module 1 — Question 9 (CO: CO1; Marks: 9)

(a) Explain how application-layer paradigms determine the distribution of services and resources in a network. (4 marks)
(b) A client retrieves a web page from www.example.test. Explain the roles of DNS and HTTP in this transaction, in the correct order. (5 marks)

Module 1 — Question 10 (CO: CO1; Marks: 9)

(a) Explain how protocol layering supports modularity and independent implementation of network functions. (4 marks)
(b) Illustrate the encapsulation sequence when a host sends application data through TCP, IP and Ethernet. Name the protocol data unit at each of these layers. (5 marks)

Module 2 — Question 11 (CO: CO2; Marks: 9)

(a) Explain how TCP provides connection-oriented service and how UDP provides connectionless service. (5 marks)
(b) A TCP receiver has received bytes 1–500 and then bytes 701–900, while bytes 501–700 are missing. State the next expected byte in cumulative acknowledgement terms and justify your answer. (4 marks)

Module 2 — Question 12 (CO: CO3; Marks: 9)

(a) Explain how I/O multiplexing supports a server that must monitor several client sockets. (4 marks)
(b) A router has these link costs: P–Q = 3, P–R = 7, Q–R = 2, Q–S = 6, R–S = 1. Compute the minimum-cost path from P to S and its total cost. (5 marks)

Module 3 — Question 13 (CO: CO4; Marks: 9)

(a) Explain the functions of the Data-Link Provider Interface and its relationship to data-link services. (5 marks)
(b) Distinguish between SOCK_PACKET and PF_PACKET as packet-socket interfaces named in the syllabus. (4 marks)

Module 3 — Question 14 (CO: CO4; Marks: 9)

(a) Explain how multiple-access protocols coordinate transmission on a shared link. (5 marks)
(b) Describe how Ethernet link-layer addresses and connecting devices support delivery within a LAN. (4 marks)

Module 4 — Question 15 (CO: CO5; Marks: 9)

(a) Explain the characteristics of transmission media and how they affect the transmission of signals. (5 marks)
(b) A channel carries 18,000 bits in 6 seconds. Compute its average data rate in bits per second. (4 marks)

Module 4 — Question 16 (CO: CO5; Marks: 9)

(a) Compare digital transmission and analog transmission in terms of signal representation and the effect of noise. (5 marks)
(b) Explain how bandwidth utilization affects the capacity and efficiency of a communication link. (4 marks)

— End of Question Paper —

B. BLUEPRINT TABLE

Q. No.	Module	Syllabus topic	CO	Bloom's level	Marks
1	1	Protocol layering; encapsulation/decapsulation	CO1	Understand	3
2	1	HTTP and FTP	CO1	Understand	3
3	2	TCP and UDP sockets	CO2	Understand	3
4	2	I/O multiplexing: select and poll	CO2	Understand	3
5	2	Network-layer protocols and routing tables	CO3	Understand	3
6	2	Intra-domain and inter-domain routing	CO3	Understand	3
7	3	Link-layer addressing; Ethernet	CO4	Understand	3
8	4	Bandwidth utilization	CO5	Understand	3
9	1	Application-layer paradigms; DNS and HTTP	CO1	Apply	9
10	1	Protocol layering; encapsulation	CO1	Apply	9
11	2	TCP/UDP; TCP cumulative acknowledgement	CO2	Apply	9
12	2	I/O multiplexing; unicast routing	CO3	Apply	9
13	3	Data-Link Provider Interface; SOCK_PACKET and PF_PACKET	CO4	Understand/Analyse	9
14	3	Multiple-access protocols; Ethernet; connecting devices	CO4	Apply	9
15	4	Transmission media; data rate	CO5	Apply	9
16	4	Digital/analog transmission; bandwidth utilization	CO5	Analyse	9

C. VERIFICATION CHECKLIST

Check	Result
Part A	8 × 3 = 24 marks
Part B	4 × 9 = 36 marks
Module distribution	Exactly two Part A questions and two Part B alternatives per module.
Part B totals	Q9–Q16 each totals 9 marks; each question has two subdivisions.
CO coverage	CO1 through CO5 assessed in the paper.
Problem-solving	Q11 protocol trace, Q12 route-cost computation, Q15 data-rate computation.
Numerical verification	Q11: cumulative ACK is 501 because bytes 501–700 are missing. Q12: P–Q–R–S cost = 3+2+1 = 6 (lower than alternatives). Q15: 18,000/6 = 3,000 bits/s.
No repetition	Questions are newly written for Set 2 and are not duplicates of Set 1; the focus and scenario differ.

Official Course Outcomes

CO1: Understand the internetworking design in terms of protocol stack and the role of various application layer protocols

CO2: Illustrate the functions of the transport layer from connectionless and connection-oriented perspectives

CO3: Identify how the network layer achieves host-to-host connectivity and caters to the diverse service requirements of the host applications

CO4: Explain the nuances of the data link layer design and demonstrate the various data link link layer protocols

CO5: Describe the fundamental characteristics of the physical layer and understand how the physical layer supports the functionalities of the top layers

D. SCHEME OF VALUATION

Part A

Q1: Encapsulation adds layer-specific headers (1); decapsulation removes them at receiver (1); supports modular, layered delivery (1).

Q2: HTTP supports web-resource transfer (1.5); FTP supports file transfer (1.5).

Q3: TCP socket supports connection-oriented byte-stream communication (1.5); UDP socket supports connectionless datagrams (1.5).

Q4: select() monitors descriptor sets (1); poll() monitors descriptor arrays (1); both allow event-driven handling of multiple descriptors (1).

Q5: Routing table stores destination/prefix and forwarding information (1); selects next hop/interface (1); supports packet forwarding (1).

Q6: Intra-domain routing operates within one routing domain (1.5); inter-domain routing exchanges routing information between domains (1.5).

Q7: Identifies local interface/device (1); enables frame source/destination identification (1); supports Ethernet frame delivery (1).

Q8: Defines use of available channel bandwidth for carrying information (1); relevant capacity/efficiency explanation (1); valid example or implication (1).

Part B

Q9(a) — 4: Explains client-server roles (2); explains distributed peer roles/resources (2).

Q9(b) — 5: DNS resolves domain name to IP address (2); client establishes HTTP request to server (1); server returns requested resource/response (2).

Q10(a) — 4: Layered division of tasks (1); defined interfaces (1); modular replacement/independence (1); implementation flexibility (1).

Q10(b) — 5: Application message (1); TCP segment (1); IP datagram (1); Ethernet frame (1); correct order/encapsulation relationship (1).

Q11(a) — 5: TCP connection setup and state (2); reliable ordered delivery (1); UDP independent datagrams without connection setup (1); overhead/service trade-off (1).

Q11(b) — 4: Identifies first missing byte as 501 (2); later bytes cannot advance cumulative acknowledgement past gap (1); cumulative ACK value 501 (1).

Q12(a) — 4: Multiple descriptors monitored (1); readiness notification (1); server processes ready sockets without blocking on each one (2).

Q12(b) — 5: Candidate path P–Q–R–S (1); sum 3+2+1 (2); total 6 (1); confirms alternatives are no cheaper (1).

Q13(a) — 5: Identifies DLPI as an interface between data-link services and users (2); explains service access/control (2); relates to data-link protocol operation (1).

Q13(b) — 4: Identifies both as packet-socket interfaces (1); SOCK_PACKET is the older interface (1); PF_PACKET is the packet-family interface (1); valid distinction in use/interface model (1).

Q14(a) — 5: Shared-link access problem (1); coordination purpose (1); access protocol operation (2); efficiency/collision consideration (1).

Q14(b) — 4: MAC addresses identify local interfaces (2); switch forwards frames within LAN (1); router connects/forwards between networks (1).

Q15(a) — 5: Names relevant medium characteristics (2); explains signal attenuation/noise or bandwidth limits (2); connects to communication quality (1).

Q15(b) — 4: Rate = bits/time (1); substitution 18,000/6 (2); answer 3,000 bits/s (1).

Q16(a) — 5: Analog representation (1); digital representation (1); analog noise effect (1); digital noise/regeneration (1); valid comparison (1).

Q16(b) — 4: Bandwidth affects available capacity (2); explains utilization/efficiency (1); identifies one consequence for communication performance (1).